bungkun1349@gmail.com
byeonggeunkim.github.io
Google Scholar | LinkedIn
Amazon AGI, Foundations-Speech & Audio

Kim, Byeonggeun

Background

I currently serve as a Senior Audio Applied Scientist at Amazon AGI with over 8 years of experience in Generative AI. As a core contributor to the Amazon Nova projects, I have extensive experience across the end-to-end audio pipeline.

I have spearheaded audio and music generation initiatives spanning audio understanding encoders, neural audio codecs, and their integration into multimodal LLMs. Currently, I am contributing to advancements in full-duplex modeling. Prior to joining Amazon, my background included deploying efficient on-device models at Qualcomm AI Research. I earned my M.S. at KAIST Korea, working with Soo-Young Lee.

Research Experience

As an Applied Scientist specializing in Audio Generative AI, my research is driven by two primary themes: enabling machines to understand and represent audio, and building systems that generate audio based on those foundations.

Amazon AGI

- Senior Applied Scientist — Seattle, WA (Apr. 2025 – Present)
- Applied Scientist II — Toronto, ON & Seattle, WA (Feb. 2023 – Mar. 2025)
  - **Audio Generation & Reconstruction:** led initiatives in audio synthesis, notably driving the integration of audio and music generation into Amazon Nova's multimodal LLMs. Currently spearheading the development of a universal audio codec designed to power seamless, full-duplex speech-to-speech interactions.
  - **Invited talks:** Yale University, CPSC 7760 Seminar — "Efficient and High Fidelity Text-to-Audio Generation: From Parallel Decoding to Causal Streaming" (Feb. 2026); Amazon Media & Entertainment ML+AI Summit — "Autoregressive text-to-audio generation approaches" (Sep. 2025).
  - **Alexa Perceptual Technology Award** (Q3/Q4 2023) — for contributions to Large Language Models raising the functional bar.

Qualcomm AI Research — Seoul, South Korea

- Staff AI Researcher (Nov. 2022 – Jan. 2023)
- Senior AI Researcher (Dec. 2019 – Nov. 2022)
- AI Researcher (Apr. 2018 – Nov. 2019)
  - **Audio Representation Learning:** focused on domain adaptation and few-shot learning for practical applications like keyword spotting and acoustic scene classification. This foundation led to my work developing the audio understanding encoder for Amazon Nova. Notable contributions include BC-ResNet and securing 1st place at the DCASE 2021 challenge through domain adaptation.
  - **1st Place, DCASE 2021 Challenge** — Task 1, Acoustic Scene Classification (Jun. 2021).

Education

Korea Advanced Institute of Science & Technology (KAIST) — Daejeon, South Korea

- M.S. in Electrical and Electronic Engineering — Feb. 2016 – Feb. 2018

Advisor: Prof. Soo-Young Lee.
Thesis: Knowledge Base Completion with Translation Based Embeddings and Negative Sampling Method.

- B.S. in Electrical and Electronic Engineering & Business Administration Feb. 2009 – Feb. 2016
  Military service: Sergeant, KATUSA (Korean Augmentation To the United States Army), 8th U.S. Army (Jun. 2010 – Mar. 2012).

Publications

(denotes equal contribution; self-name in bold.)*

1. **Interspeech 2026 (Under Review)** *Improving Text-to-Audio Instruction Following via Fine-Grained Feedback from Audio-Aware Large Language Models*.
2. **NeurIPS 2026 (Under Review)** *Fast Text-to-Audio Generation with One-Step Sampling via Energy-Scoring and Auxiliary Contextual Representation Distillation*.
3. **ICML 2025** *Generative Audio Language Modeling with Continuous-valued Tokens and Masked Next-Token Prediction* [PDF]
   Shu-wen Yang, **Byeonggeun Kim**, Kuan-Po Huang, Qingming Tang, Huy Phan, Bo-Ru Lu, et al. *(mentored internship)*
4. **ICML 2025** *IMPACT: Iterative Mask-based Parallel Decoding for Text-to-Audio Generation with Diffusion Modeling* [PDF]
   Kuan-Po Huang, Shu-wen Yang, Huy Phan, Bo-Ru Lu, **Byeonggeun Kim**, et al. *(mentored internship)*
5. **ICASSP 2025 (Oral)** *Effective Techniques for Scaling Audio Encoder Pretraining* [PDF]
   Byeonggeun Kim*, Andrew Bydlon*, Qingming Tang, Huy Phan, Chieh-Chi Kao, Tao Chang, Chao Wang.
6. **NeurIPS 2025 Workshop** *Unlocking Transfer Learning for Open-World Few-Shot Recognition* [PDF]
   Byeonggeun Kim*, Jun-Tae Lee*, Kyuhong Shim, Simyung Chang. *(Reliable ML Workshop)*
7. **Amazon Tech Report 2025** *Amazon Nova 2: Multimodal reasoning and generation models* [PDF]
   Amazon AGI *(contributor: audio/music generation, audio encoder & codec)*.
8. **ICASSP 2024** *Cross-Triggering Issue in Audio Event Detection and Mitigation* [PDF]
   Huy Phan, **Byeonggeun Kim**, Vu Nguyen, Andrew Bydlon, Qingming Tang, et al.
9. **ICIP 2023** *Task-Agnostic Open-Set Prototype for Few-Shot Open-Set Recognition* [PDF]
   Byeonggeun Kim*, Jun-Tae Lee*, Kyuhong Shim, Simyung Chang.
10. **ICASSP 2023 (Oral)** *Scalable Weight Reparametrization for Efficient Transfer Learning* [PDF]
    Byeonggeun Kim*, Jun-Tae Lee*, Seunghan Yang, Simyung Chang.
11. **Interspeech 2023 (Oral)** *Improving Small Footprint Few-shot Keyword Spotting with Supervision on Auxiliary Data* [PDF]
    Seunghan Yang, **Byeonggeun Kim**, Kyuhong Shim, Simyung Chang.
12. **ICLR 2023** *TTN: A Domain-Shift Aware Batch Normalization in Test-Time Adaptation* [PDF]
    Hyesu Lim, **Byeonggeun Kim**, Jaegul Choo, Sungha Choi. *(mentored internship)*

13. Interspeech 2022 (Oral) *Personalized Keyword Spotting through Multi-task Learning* [PDF]
Seunghan Yang, **Byeonggeun Kim**, Inseop Chung, Simyung Chang.
14. Interspeech 2022 *Dummy Prototypical Networks for Few-shot Open-set Keyword Spotting* [PDF]
Byeonggeun Kim, Seunghan Yang, Inseop Chung, Simyung Chang.
15. Interspeech 2022 *Domain Generalization with Relaxed Instance Frequency-wise Normalization for Multi-device Acoustic Scene Classification* [PDF]
Byeonggeun Kim, Seunghan Yang, Jangho Kim, Hyunsin Park, Juntae Lee, Simyung Chang.
16. Interspeech 2021 *Broadcasted Residual Learning for Efficient Keyword Spotting* (BC-ResNet) [PDF] [Code]
Byeonggeun Kim*, Simyung Chang*, Jinkyu Lee, Dooyong Sung.
17. DCASE 2021 Workshop *Domain Generalization on Efficient Acoustic Scene Classification Using Residual Normalization* [PDF]
Byeonggeun Kim, Seunghan Yang, Jangho Kim, Simyung Chang.
18. DCASE 2021 Challenge — 1st Place *QTI Submission to DCASE 2021: Residual Normalization for Device-Imbalanced Acoustic Scene Classification with Efficient Design* [PDF]
Byeonggeun Kim, Seunghan Yang, Jangho Kim, Simyung Chang.
19. ASRU 2019 *Query-by-Example On-Device Keyword Spotting* [PDF] [Dataset]
Byeonggeun Kim, Mingu Lee, Jinkyu Lee, Yeonseok Kim, Kyuwoong Hwang.
20. ASRU 2019 *Orthogonality Constrained Multi-Head Attention for Keyword Spotting* [PDF]
Mingu Lee, Jinkyu Lee, Hye Jin Jang, **Byeonggeun Kim**, Wonil Chang, Kyuwoong Hwang.

Granted Patents

1. *Multi-task learning for personalized keyword spotting* [PDF] US 12,347,439
Seunghan Yang, **Byeonggeun Kim**, Inseop Chung, Simyung Chang. *(Jul. 2025)*
2. *Relaxed instance frequency normalization for neural-network-based audio processing* [PDF] US 12,266,379
Byeonggeun Kim, Seunghan Yang, Hyunsin Park, Juntae Lee, Simyung Chang. *(Apr. 2025)*
3. *Target Keyword Selection* [PDF] US 12,039,969
Wonil Chang, Jinseok Lee, . . . , **Byeonggeun Kim**, et al. *(Jul. 2024)*
4. *Task Agnostic Open-set Prototypes for Few-shot Open-set Recognition* [PDF] US 12,019,641
Byeonggeun Kim, Juntae Lee, Simyung Chang. *(Jun. 2024)*
5. *Systems and methods of image processing based on gaze detection* [PDF] US 11,798,204
Hyunsin Park, Juntae Lee, Simyung Chang, **Byeonggeun Kim**, Jaewon Choi, Kyu Woong Hwang. *(Oct. 2023)*
6. *On-device self training in two-stage wakeup system. . .* [PDF] US 11,664,012
Young Mo Kang, Sungrack Yun, Kyu Woong Hwang, Hye Jin Jang, **Byeonggeun**

Kim. *(May 2023)*

7. *Activating speech recognition based on hand patterns detected using plurality of filters* [PDF] US 11,437,031
   Sungrack Yun, Young Mo Kang, Hye Jin Jang, **Byeonggeun Kim**, Kyu Woong Hwang. *(Sep. 2022)*
8. *Method and apparatus for activating speech recognition* [PDF] US 11,205,433
   Byeonggeun Kim, Young Mo Kang, Sungrack Yun, Kyu Woong Hwang, Hye Jin Jang. *(Dec. 2021)*

Professional Activities

Area Chair: ICASSP 2026.
Session Chair: INTERSPEECH 2022.
Reviewer: ICASSP (2023–2026), INTERSPEECH (2023–2026), CVPR (2024–2026), ICCV 2025, ECCV 2026, ACL 2025, AAAI (2025–2026), WACV (2025, 2026), BMVC 2026, NeurIPS 2026, EMNLP 2025.

Languages

Korean (native), English (full professional proficiency), Japanese (conversational).